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Roll No.

220944

4th Sem / Electrical

**Subject : Estimating and Costing in
Electrical Engineering**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The generating voltage at generating station is step up for transmission is _____

- a) 765 kv b) 66kv
- c) 1200kv d) 11kv

Q.2 The voltage between any two phases of a 400 v 3 phase 3 wire system is

- a) 220 V b) 110V
- c) 250 V d) 400 V

Q.3 Which among these is a demerit of underground service main?

- a) Ugly appearance
- b) Frequent fault occurrence
- c) Costly d) None of these

- Q.4 According to National Electric Code. the colour code for the neutral is
- a) Green
 - b) Blue
 - c) Red
 - d) Black
- Q.5 Which device used for excess current protection?
- a) TPIC
 - b) DPIC main switch
 - c) Isolator
 - d) Miniature circuit Breaker
- Q.6 Earnest money is generally asked to be deposited
- a) at the time of purchase tender document
 - b) at the time of entering the agreement
 - c) by the successful bidder after he gets the letter of acceptance
 - d) at the time of submission of bid

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define supervision charges.
- Q.8 What is the full form of DPDT?
- Q.9 Expand MCB?
- Q.10 Define earthing?
- Q.11 What is a Isolator?
- Q.12 What is net price?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What do you mean by extension of data of tender?
- Q.14 What is tender? Discuss about tender notice.
- Q.15 What are the main points considered for conduit wiring?
- Q.16 What is the difference between limited tender and global tender?
- Q.17 Why it is necessary to provide earthing in a domestic electric installation?
- Q.18 Discuss overhead method of service connection.
- Q.19 Explain labour cost? Also name the various methods calculating labour charges.
- Q.20 Calculate the full load current and starting current for a 10 HP, 3 phase, 400 Volts Induction motor, power factor 0.85 lagging.
- Q.21 Explain the purpose of comparative statement.
- Q.22 What is price list, explain with example.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 The plan of a two rooms house with verandah and a kitchen is given below draw the installation plan and estimate the quantity of material required. The

wiring is to be done in concealed conduit system and assume the supply voltage is 220 volt, single phase the details of electrical point is as under

	Lamp 100 watt	Tube 40 watt	Fan 80 watt	5 amp Socket
Verandah	1			
Room No. 1	1	1	1	1
Room No. 2	1	1	1	1

Room No. 1 5m*5m	Room No. 2 5m*5m	Kitchen 2m*3m
		Verandah 3m*3m

Q.24 Prepare a tender notice performa for the purchasing of 100 numbers of 11kv/415 volts, 3 phase, 50 Hz. 200 kVA transformer.

Q.25 Prepare a list of material for a service connective by cable to a house (single store) at 230 volts single phase 50 Hz,. The size of the cable is 10 sq mm aluminium. The supply is to be given from over head 415 volts 3 phase 4 wire line which is 20 meters away from the building.